

Challenge 1: Warehouse Spatial Intelligence

Team: 3CE_team

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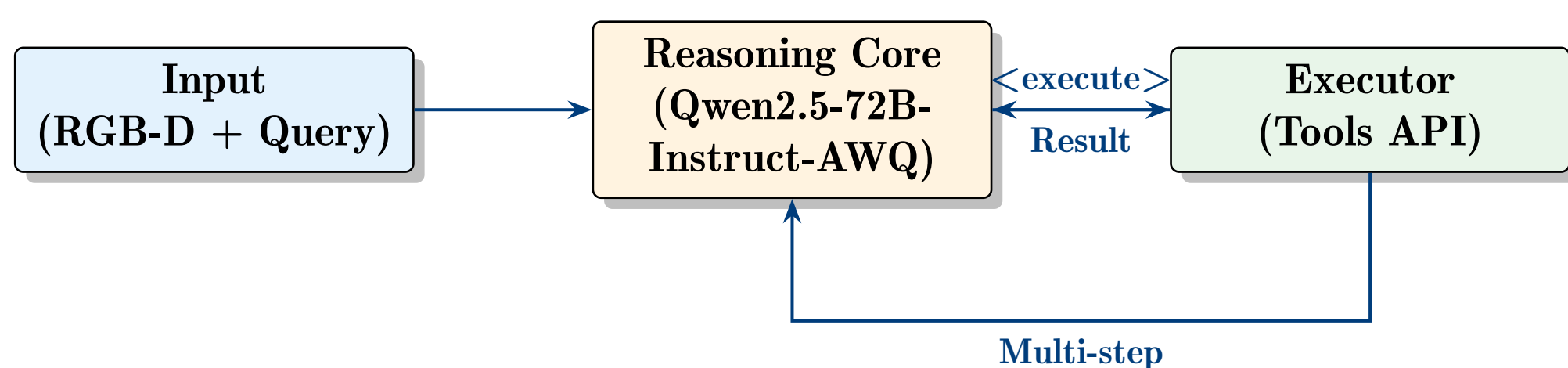
Task Overview

Answer fine-grained spatial questions grounded on multiple specific regions in dense warehouse scenes.

- **Input:** RGB image, Depth map, RLE masks dictionary, Question.
- **Goal:** Generate normalized answers (distance in meters, left/right relation, region-id selection, and counts).
- **Challenge:** Dense objects, depth noise, and invalid region references in free-form reasoning.

System Architecture

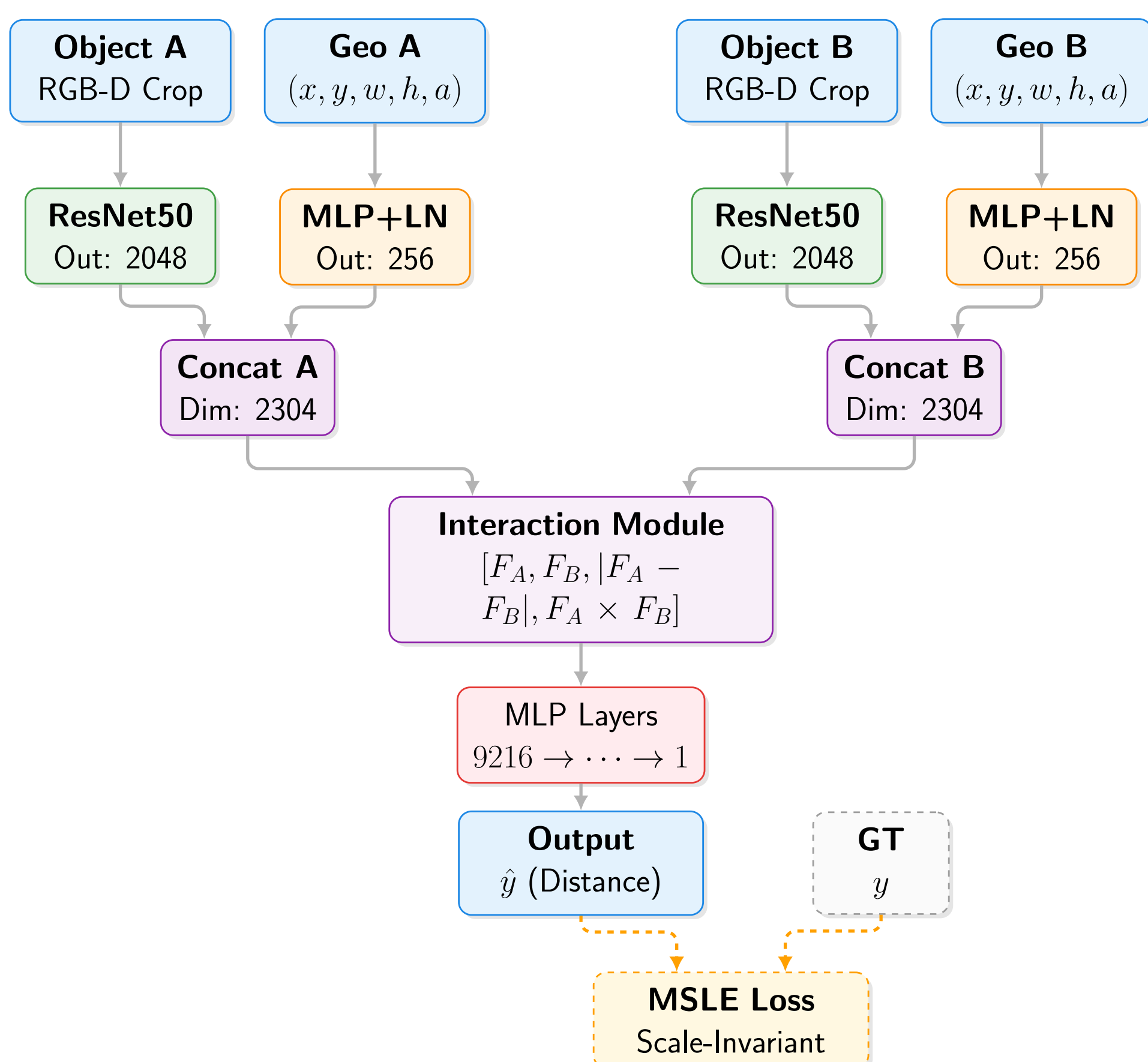
We employ a Tool-Augmented Agent using Qwen2.5-72B-Instruct-AWQ as the reasoning core. The LLM proposes tool calls and the executor returns computed results.



DistanceTool (Siamese RGB-D Regression)

LLMs are unreliable for metric distance estimation; we therefore use a dedicated Siamese RGB-D regressor.

- **Geometric Fusion:** combines RGB-D crop features with a bbox-based geometry vector $(c_x, c_y, w, h, w \times h)$ (normalized by image size).
- **Loss Function:** MSLE loss $\mathcal{L} = \text{MSE}(\ln(1 + \hat{y}), \ln(1 + y))$.
- **Logic:** if either region is a **buffer** and $\text{inside}(\text{buffer}, [\text{obj}])=1$, we override the distance to 0.0.



InsideTool (Geometry-Aware Classification)

Containment prediction between a container region and candidate object regions.

- **Input Construction:** 5-channel tensor with explicit spatial context:

$$\text{Input} = \text{Concat}(\text{RGB}_{[3]}, \text{Mask}_{\text{Container}[1]}, \text{Mask}_{\text{Object}[1]})$$

- **Preprocessing:** masks are scaled $(0 \rightarrow 255)$, bilinearly resized to $(360, 640)$, then normalized back to $[0, 1]$.
- **Counting:** `inside(container, list_of_objs)` runs batch inference and returns the number of positives (single candidate yields 0/1).

Reasoning Trace Example

Example of multi-step tool use with `<execute>` and `<answer>`.

User: "Is the pallet closer to the buffer or the forklift?"

Agent: I need to measure distances.

`<execute>dist(pallet_1, buffer_1)</execute>`

Tool Output: 2.45

`<execute>dist(pallet_1, forklift_1)</execute>`

Tool Output: 5.12

Agent: Select the smaller distance.

`<answer>buffer_1</answer>`

Deterministic Utilities

Rule-based utilities for stable spatial relations:

- **Left/Right:** compare mask centroids $(c_x^A < c_x^B)$.
- **Closest:** batch distance evaluation and `argmin(closest)`.
- **Is Empty:** for each transporter, compute its maximum IoU over all pallets, then select the transporter with the smallest max-IoU (minimax overlap).

Results

Model	Validation					Test
	Count	Dist.	L/R	MCQ	Score	
Qwen2.5-14B	<u>79.25</u>	<u>90.67</u>	<u>100.00</u>	<u>88.59</u>	<u>89.63</u>	<u>82.48</u>
Qwen2.5-72B-Instruct-AWQ	93.08	97.33	100.00	90.60	95.25	95.65

Conclusion

We couple an instruction-tuned LLM with explicit spatial tools under a constrained execution interface.

- **Tool-grounded numerics:** distance and containment are computed by dedicated models (`dist/inside`) when invoked, instead of free-form generation.
- **Constrained execution:** a function whitelist and structured argument parsing reject malformed tool calls; failed items return `-1` rather than fabricated outputs.